Vibrational energy harvesting in micro-electro-mechanical systems integrated with sacrificial anode metal sheets for detecting steel corrosion in concrete

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Abstract. Reinforced concrete (RC) structures in marine environments are highly susceptible to deterioration caused by chloride-induced corrosion of reinforcing steel bars. Although substantial resources are often allocated for maintenance, developing more efficient systems for monitoring RC structures in corrosive conditions is essential to ensure their long-term durability and safety. This research aims to establish a clear connection between the surrounding corrosion environment and the behaviour of steel bars embedded in concrete, which is vital for maintaining the required service level of RC structures. Sacrificial Anode Metal Sheets (SAMS), integrated with a Micro Energy Harvester (MEH) based on Micro-Electro-Mechanical Systems (MEMS) technology, are proposed and employed to continuously monitor the environmental conditions affecting the degradation of the targeted structure. The sensor detects the frequency shift of vibrational behaviour in real-time, providing critical information on corrosion progress of steel bars in concrete. To modify the monitoring accuracy, Half-Cell Potential (HCP) measurement for rebars in concrete is conducted to correlate the corrosion of SAMS with the steel bars inside the concrete. Frequency changes detected by MEMS-SAMS strongly correlate to HCP data, confirming that MEMS-SAMS can effectively monitor the corrosion due to chloride attack. This relationship significantly enhances early detection and optimizes maintenance strategies for RC structures in highly corrosive environments.

1 Introduction

1.1 Background

Reinforced concrete is a fundamental material in modern infrastructure, known for its strength and durability. However, one of the biggest challenges in maintaining these structures is the corrosion of steel reinforcement (rebars) when exposed to harsh environmental conditions such as marine, industrial, and urban areas [1]. Corrosion weakens concrete, causing cracks, spalling, and, in severe cases, structural failure. This not only leads to costly repairs but also raises serious safety concerns. The main cause of rebar corrosion is chloride, either by ingress, which can happen due to exposure to seawater or de-icing salts, or by contaminated construction materials [2]. When chlorides penetrate the concrete cover, or are present in the cover zone, and reach the steel, they break down the protective oxide layer, initiating corrosion. This often leads to pitting corrosion, the localized and aggressive form of deterioration that can significantly degrade the rebar. This process accelerates even further when the concrete cracks, allowing moisture and oxygen to reach the rebar directly [3]. Additionally, factors like high humidity, carbonation, and repeated wet-dry cycles contribute to corrosion, making structures in coastal and humid regions particularly vulnerable.

In Japan, where many bridges, tunnels, and marine structures are exposed to these harsh conditions, corrosion is a major concern [4]. To address this, researchers have developed advanced monitoring systems to detect and prevent deterioration before it compromises structural integrity. Despite various protective measures, predicting when and where corrosion will occur remains a challenge. Common inspection methods, such as visual checks and electrochemical tests [5], often detect damage only after it has already progressed. This highlights the need for more advanced monitoring systems that can provide real-time data on corrosion activity remotely and non-destructively. Since corrosion significantly affects the long-term durability of concrete structures, integrating predictive maintenance strategies is crucial. Life-Cycle Management (LCM) ensures that durability design and long-term maintenance strategies are aligned to sustain the structural performance of concrete throughout its service life [6].

To overcome these challenges, researchers are turning to real-time, data-driven approaches. Structural Health Monitoring (SHM) systems provide an effective solution to reduce the time, cost, and labour required for traditional inspection methods [7]. With the rapid advancement of the Internet of Things (IoT) technology, smart sensors and connected networks are increasingly being integrated into civil infrastructure [8]. These systems are particularly valuable for monitoring bridges and other critical structures, allowing for continuous data collection and early detection of structural changes.

To further enhance efficiency and sustainability, vibrational energy harvesters have gained attention as a self-powered solution for IoT and SHM systems. These devices can convert mechanical vibrations from the structure into electrical energy, eliminating the need for external power sources. Among these, Micro-Electro-Mechanical Systems (MEMS)-based micro energy harvesters (MEH) are a promising option due to their compact size, low power consumption, and ability to function autonomously [9,10].

Addressing the challenge of corrosion monitoring, MEMS integrated with Sacrificial Anode Metal Sheets (MEMS-SAMS) offer an innovative approach. By detecting frequency shifts caused by corrosion-induced degradation at SAMS, MEMS-SAMS technology monitors the environmental effects on corrosion progression of rebar in concrete, offering insights that contribute to structural health assessment. In addition to real-time monitoring, ensuring a self-sustained power source is essential for continuous operation. To address this, vibrational energy harvesters have gained attention as a self-powered solution. This enables proactive maintenance, reduces long-term repair costs, and extends the lifespan of reinforced concrete structures. With the combination of IoT-driven SHM systems, energy-harvesting sensors, and MEMS-SAMS technology, structural and environmental monitoring for RC structures can become more efficient, cost-effective, and sustainable. These advancements are critical for ensuring the durability and safety of civil structures in challenging environments.

1.2 Development of MEMS monitoring sensors

1.1.1 Overview

A vibration-based micro energy harvester (MEH) device is proposed for monitoring environmental and structural conditions in reinforced concrete (RC) structures. In this study, the MEH device is developed using Micro-Electro-Mechanical Systems (MEMS) technology [11]. These devices harvest micro-energy from environmental vibrations, converting it into electrical energy. The ability of vibration-based MEH to operate autonomously makes it a suitable solution for various Structural Health Monitoring (SHM) applications in civil infrastructure. However, beyond general SHM, the energy harvested by the MEH system responds to changes in vibrational characteristics, which may indicate structural alterations such as material degradation or environmental influences [12]. This makes the MEH device capable of providing real-time insights into both structural health and environmental stressors. The proposed device is based on a previous study [13], which introduced an inexpensive, coin-sized energy harvester capable of generating over 100 μW RMS of electrical power under 0.03 G_{RMS} accelerations. By utilizing ambient vibrations as an energy source, the MEH device eliminates the need for external power, making it a self-sustaining sensor ideal for long-term monitoring in remote or inaccessible locations [14].

The MEMS sensor used in this study is compact, measuring 23 × 23 mm, as shown in Figure 1. This small form factor provides several advantages. The compact design reduces manufacturing and packaging costs, enabling large-scale production at a lower price. Additionally, the wireless nature of the sensor allows for flexible deployment in various locations without the need for complex wiring. These features make the sensor highly suitable for large-scale monitoring network for bridges and other critical infrastructure.

The MEMS sensor operates as a comb-shaped silicon electret, designed to convert mechanical vibrations into electrical energy. The energy conversion process is based on the interaction between movable and fixed comb electrodes when the cantilever vibrates out of the plane [15]. As the cantilever vibrates, the overlapping area between the movable and fixed comb electrodes changes, altering the capacitance between the electrodes and affecting the electric field within the device. These fluctuations induce electron movement in the fixed comb electrodes, generating an electric current. Importantly, the electrical output of the MEH device can be directly influenced by changes in the dominant frequency and amplitude of vibrations. As environmental conditions or material properties change, shifts in the dominant frequency affect the efficiency of energy harvesting. This makes the MEMS-based MEH a valuable tool for real-time monitoring of vibrational behaviour, enabling its use in detecting structural and environmental changes in RC infrastructure.

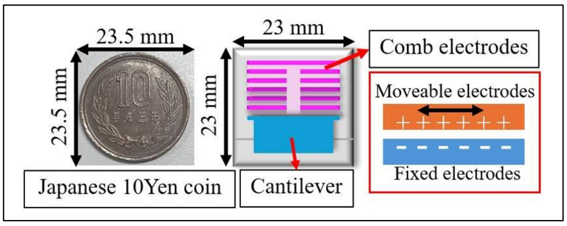

Fig. 1. A model of vibration-based micro energy harvester (MEH) (reproduced from [13]).

The efficiency of this process depends on several factors, including the precision of the MEMS design, the material properties of the electrodes, and external environmental conditions. With its ability to efficiently convert mechanical vibrations into electrical energy, the MEMS-based energy harvester is highly valuable for both energy harvesting and real-time structural health monitoring. This innovative approach to self-powered monitoring systems ensures reliable and continuous assessment of structural conditions, helping to detect early signs of deterioration and reducing maintenance costs for critical infrastructure.

1.1.2 Energy harvesting calculations

The vibration power generation device, specifically a MEMS vibrational power generator, consists of a fixed and a movable comb electrode. When the movable mass of the harvester oscillates at its resonance frequency, the expected root mean square (RMS) output power, P, can be determined using Equations (1) and (2) [13]. Presuming that the internal impedance of the vibration power generator matches the connected device, the expected RMS output power can be calculated while considering the conversion efficiency from the linear equivalent circuit of the electrostatic actuator. This ensures an accurate assessment of the energy harvesting capability under specific operating conditions.

$$P = \frac{ma^2Q}{4\omega_0}\eta \quad (1)$$

$$\eta = \frac{1}{1+(\frac{kC_0}{QA^2})^2} \quad (2)$$

The power generation amount, P, is measured in watts (W) and is influenced by several key parameters. These include m, the mass of the movable body in kilograms (kg), and a, the effective acceleration of the movable body in meters per second squared (m/s²). The quality factor, Q, is a dimensionless parameter that characterizes the damping of the system, while η represents the impedance matching ratio, which determines the proportion of output power relative to the total power. Additionally, C_0 denotes the electrostatic capacitance, and A represents the force coefficient, measured in coulombs per meter (C/m). This coefficient affects the power efficiency of vibrational power generator. The angular resonance frequency, ω_0, is expressed in radians per second (rad/s) and plays a crucial role in energy conversion efficiency. For optimal energy harvesting performance, the angular frequency of the harvester is typically tuned to match the resonance frequency of incoming vibrations, ensuring efficient kinetic-to-electrical energy conversion [16]. Table 1 presents the design parameters of the MEMS vibrational power generator analyzed in this study.

Table 1. MEH structural and material properties.

Parameter (unit)	Value
B (nF/m): Comb Coefficient	70.5
C_1 (pF): Stray Capacitance	68
k_x (N/m): Spring constant	730
V_0 (V): Charged Voltage	250
R (MΩ): External Load Resistance	5
m (mg): Moveable Body Mass	74
w_1 (μm): Length of Comb	700

1.1.3 MEMS-SAMS Design to Monitor Corrosion

MEMS devices have been effectively applied in real-world scenarios, proving their practicality for structural monitoring [11,12]. The vibrational Micro Energy Harvester (MEH) is designed to detect variations in modal frequency and amplitude within targeted structures. Expanding on this concept, previous studies introduced the incorporation of the Sacrificial Anode Metal Sheet (SAMS) into MEMS to identify chloride-rich environments [17], as shown in Figure 2. SAMS is a corrosion-prone metal sheet selected for its reactivity in aggressive environments, aiming to simulate key aspects of steel reinforcement corrosion in concrete. This approach relies on tracking the dominant frequency and amplitude in the metal sheets, which signals corrosion progression and impacts the device's energy harvesting capabilities. As corrosion progresses, SAMS undergoes material degradation, altering its vibrational behaviour and causing shifts in its dominant frequency. These frequency shifts, which is caused by the corrosion of SAMS cantilever, directly affect the interaction between the movable and fixed comb electrodes within the MEH device. This leads to variations in capacitance and modifies the electric field, ultimately influencing the device's electrical output. The resulting electron movement within the fixed comb electrodes generates an electrical signal that provides real-time insight into environmental and corrosion conditions.

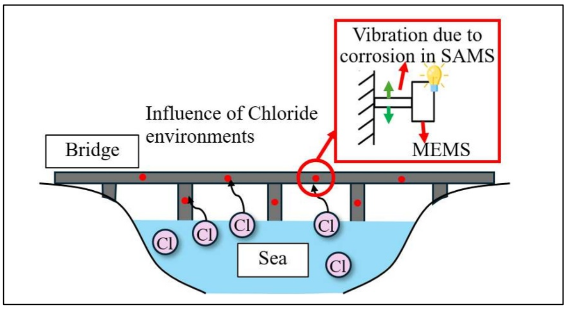

Fig. 2. MEMS-SAMS device to monitor corrosion [17].

The MEMS-SAMS system operates by connecting to a circuit with resistance components, where LEDs indicate voltage changes. Corrosion in reinforced concrete structures is indirectly assessed by monitoring the frequency shift of SAMS. Since SAMS is exposed to the same chloride-rich environment as the targeted concrete structure with reinforcement embedded, the corrosion-induced frequency drop of SAMS serves as the indicator of conditions that promote rebar degradation. When the frequency reaches a predefined threshold for electrical output according to the design of MEH device, it emits the signal corresponding to corrosion inside the concrete, as presented in Figure 3. The system responds to environmental changes affecting the steel bar, with higher frequencies and greater amplitudes resulting in increased electrical output. By analyzing the power generation based on the initial characteristics of SAMS, such as the original vibrational frequency and amplitude, the system effectively identifies external environmental conditions that contribute to steel corrosion inside the concrete. This capability enables early detection and supports planning maintenance strategies to mitigate structural deterioration.

A key limitation of previous studies has been the uncertainty regarding whether MEMS-SAMS alone can accurately predict corrosion occurring within the structure. To address this issue, the present study aims to establish a clear relationship between external environmental conditions detected by MEMS-SAMS and the internal corrosion state of reinforcing steel. This is accomplished through controlled laboratory testing, where Half-Cell Potential (HCP) measurements are used to validate the effectiveness of MEMS-SAMS in monitoring corrosion progression. By correlating these findings, this study enhances the credibility of MEMS-SAMS technology before its future deployment in real-world structures.

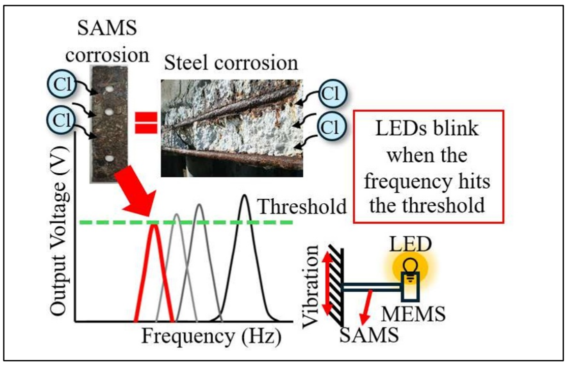

Fig. 3. MEMS-SAMS device's model: Frequency shifts in SAMS due to corrosion reflect the progression of concrete corrosion [17].

The Half-Cell Potential (HCP) method is one of the most established and widely utilized techniques for detecting corrosion in reinforced concrete structures. With a well-developed theoretical foundation and a strong track record of reliability, HCP measurements are extensively applied in both laboratory research and field engineering to assess the electrochemical state of embedded reinforcement [18,19]. This method provides valuable insights into corrosion activity by measuring the electrical potential of steel rebars within the concrete, helping engineers determine the likelihood and extent of corrosion. However, it is important to note that in fully saturated concrete, Half-Cell Potential (HCP) readings may be high even when corrosion is not actively occurring due to limited oxygen availability, which can affect interpretation.

In this study, HCP measurements were conducted under controlled environmental conditions to assess the effectiveness of MEMS-SAMS for corrosion monitoring. The recorded HCP trends over time were systematically analyzed and compared with previously obtained MEMS-SAMS frequency degradation data. This comparative approach was crucial in validating the capability of MEMS-SAMS to detect external environmental changes and assess their impact on internal corrosion progression. By correlating HCP measurements with MEMS-SAMS frequency shifts, this study aimed to establish a direct link between external corrosion-inducing factors and their effects on reinforcement deterioration. The findings contribute to the advancement of real-time monitoring technologies, ensuring a more proactive approach to corrosion detection and infrastructure maintenance.

1.2 Objectives

This study aims to establish a correlation between MEMS-SAMS data and Half-Cell Potential (HCP) measurements under controlled laboratory conditions, validating the capability of MEMS-SAMS to detect corrosion-related changes in the surrounding environment. Additionally, it seeks to evaluate MEMS-SAMS as a corrosion monitoring system by analyzing its effectiveness in detecting environmental variations through frequency shifts and electrical output. This evaluation will help determine whether monitoring only the external environment can reliably predict corrosion progression within reinforced concrete structures. By achieving these objectives, the research aims to enhance the reliability of MEMS-SAMS for real-world corrosion detection and structural health monitoring.

2 Materials and Methods

2.1. Materials and specimen design

The experimental setup involved casting reinforced concrete specimens with dimensions of 16 × 4 × 4 cm. Each specimen contained a 10 mm diameter steel bar, embedded at a depth of 2.5 cm from the concrete surface to the bar center. The concrete mix design, summarized in Table 2, was prepared with a water-cement ratio of 0.6 to facilitate proper workability and accelerate testing. The coarse aggregate used had a maximum gravel size of 13 mm. After casting, the specimens were cured under standard laboratory conditions following ASTM C192, ensuring proper hydration and strength development before testing.

Table 2. Mix design.

Material	Total
Cement (kg/m^3)	336
Water (kg/m^3)	201
Sand (kg/m^3)	918
Gravel (kg/m^3)	902

To enhance the reliability of Half-Cell Potential (HCP) measurements and minimize external influences, all surfaces of the specimen were coated with epoxy resin, except for one designated measurement side, which remained uncoated. This selective coating, as illustrated in Figure 4, helped isolate the measurement zones, reducing the impact of environmental factors and improving the accuracy and consistency of the experimental results.

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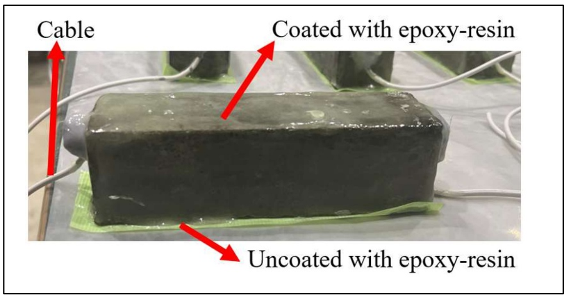

Fig. 4. Epoxy Coating for HCP Measurements.

Following the curing process, the specimens were exposed to six different environmental conditions, with two specimens per condition, as outlined in Table 3. These conditions were carefully selected based on previous MEMS-SAMS results to represent a wide range of possible corrosion scenarios. The experimental setup maintained a constant temperature of 20°C and 40°C inside a closed chamber with 95% relative humidity, simulating an aggressive environment that could accelerate the corrosion process. To investigate the effects of different corrosion-inducing factors, both chloride-rich (NaCl solution) and chloride-free (deionized water) environments were included. Each condition was designed to replicate real-world exposure scenarios that reinforced concrete structures might encounter, particularly in marine and high-humidity environments.

Table 3. Exposure conditions of environmental scenarios.

Solution	Temperature	Condition
3% NaCl (N)	20ºC	Wet-Dry (D)
	40ºC	Wet-Dry (D)
		Submerged (S)
Deionized water (H)	20ºC	Wet-Dry (D)
	40ºC	Wet-Dry (D)
		Submerged (S)

For the wet and dry (W/D) conditions, a fabric sheet was placed over the specimens and periodically soaked with the designated solution. The solution was applied every three days to create a consistent wet-dry cycle, ensuring periodic moisture exposure. Instead of directly pouring water on the specimens for three days and leaving them without water for four days, the fabric sheet method was chosen. This approach was based on findings from previous studies on the frequency degradation of metal sheets, where fabric sheets were used to maintain consistency in wet-dry cycles. By applying the same method, potential discrepancies in moisture exposure between experiments were minimized, improving the reliability of results.

For the submerged conditions, the specimens were fully immersed in their respective solutions, ensuring continuous exposure to either NaCl solution or deionized water. The setup of wet and dry and submerged conditions, illustrated in Figure 5, allowed for direct comparison between fluctuating moisture conditions (wet-dry cycles) and constant immersion, helping to evaluate their effects on corrosion progression and frequency degradation.

Fig. 5. The concrete specimen in wet and dry condition (left) and submerged condition (right).

2.2 Corrosion test

The half-cell potential (HCP) test was employed to measure the corrosion potential of the specimens non-destructively, ensuring that the concrete remained intact while assessing the likelihood of corrosion. A Silver-Silver Chloride (Ag/AgCl) reference electrode was used to measure the potential of the steel reinforcement. The recorded values were adjusted by subtracting 90 mV to convert them to an equivalent Copper/copper sulfate (CSE), in line with ASTM C876 for standardization and comparison. Since ASTM C876 is based on $Cu/CuSO_4$ reference values, this conversion ensures consistency with the classification criteria outlined in Table 4, as the experimental temperature remained around 20°C.

Table 4. Corrosion probability according to ASTM C876.

HCP (CSE) / mV	Probability
E > -200	90% no corrosion
-350 < E < -200	Uncertain
E > -350	90% corrosion

HCP measurements were conducted every two weeks over 24 weeks to monitor the progression of corrosion. To ensure accurate and consistent measurements, the specimens were pre-soaked in water for one hour before each reading to ensure consistent surface moisture. This step was necessary to moisten the surface, which improved electrical contact between the reference electrode and the concrete, minimizing measurement fluctuations. Each specimen was tested at two designated locations above the surface, with measurements taken at a single cable connection point per specimen. This setup was designed to enhance accuracy and reduce variability, ensuring that the recorded values reflected the true corrosion potential across the monitored regions. These measurements provided valuable insights into the corrosion behavior of the reinforcing steel under different environmental conditions, contributing to a better understanding of how moisture and chloride exposure impact the deterioration process over time.

3 RESULTS AND DISCUSSION

3.1 The relationship between HCP results and Frequency degradation results

During the curing period, all specimens were immersed in water for 28 days to ensure proper hydration. Throughout this period, the half-cell potential (HCP) of the steel bars remained relatively stable, ranging between -200 and -120 mV vs. CSE, which falls within the 90% no-corrosion condition based on ASTM C876. This stability was expected, as the steel bars were new and had not been exposed to extreme environmental conditions. Additionally, the protective coating applied as a corrosion prevention method did not significantly affect the HCP during the curing phase, as the concrete remained sufficiently permeable to moisture and ions, allowing normal electrochemical stabilization.

After the 28-day curing period, the specimens were exposed to different environmental conditions for two weeks before measurements began. At the start of the exposure test, the HCP values for the wet and dry conditions at both 20°C and 40°C remained within the 90% no-corrosion range. However, for the submerged condition, the initial readings were already lower than -250 mV vs. CSE, suggesting early signs of electrochemical activity. The observed changes could be attributed to oxygen starvation in submerged conditions, which limits cathodic reactions. Other contributing factors may include increased electrolyte conductivity or minor passive film breakdown on the steel surface. Additionally, the transition from curing to exposure could have temporarily disrupted the electrochemical equilibrium, influencing the initial readings of HCP.

To minimize potential measurement errors, we focused on analyzing the trend of HCP reduction over time rather than absolute initial values. This approach allows for a clearer understanding of how corrosion progresses under different environmental conditions and helps differentiate between temporary fluctuations and significant degradation trends.

In this study, the frequency degradation of the SAMS measured at 20 mm from the previous research [15] is used for comparison. Figures 6 to 8 illustrate the relationship between the frequency drop of SAMS and Half-Cell Potential (HCP) of the steel bar in concrete under the same exposure conditions. The results indicate a consistent decline in both frequency degradation and HCP over time, confirming that corrosion of the steel bar affects both the mechanical properties (observed through the frequency drop in MEMS-SAMS) and the electrochemical properties (measured by HCP values). The rate of decline varies depending on environmental conditions, highlighting the influence of chloride and moisture exposure on corrosion progression inside the concrete.

Notably, in 20°C and 40°C NaCl under wet and dry conditions, both frequency degradation and HCP exhibit a sharp decline, as shown in Figure 6. A closer examination of Figure 6 (a) reveals that HCP and frequency drop follow a similar trend until week 10, after which HCP temporarily increases before resuming its decline, while frequency drop continues decreasing. HCP temporarily rises due to rust-induced passivation but declines as chloride ingress disrupts the layer, accelerating corrosion progression. In Figure 6(b), the trends remain closely aligned throughout the period, further validating the potential of MEMS-SAMS to detect environmental changes that influence corrosion progression in concrete structures.

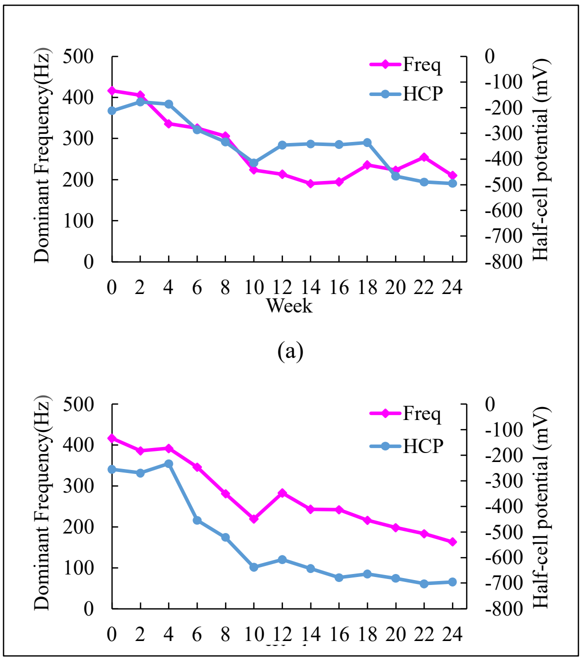

Fig. 6. Dominant frequency degradation and half-cell potential over time for NaCl solution in wet and dry conditions: (a) 20°C and (b) 40°C.

Figure 7 illustrates the dominant frequency degradation trend and Half-Cell Potential (HCP) variation over time in deionized water under wet and dry conditions at (a) 20°C and (b) 40°C. In both conditions, the dominant frequency and HCP exhibit a gradual decline, reflecting electrochemical changes within the system, such as increased moisture content or shifts in the steel's passive layer stability. The frequency drop reflects the weakening mechanical integrity of the SAMS, while the decreasing HCP suggests increasing electrochemical activity associated with corrosion progression. At 20°C (Figure 7 (a)), the HCP initially remains stable but begins to decline significantly after week 6, followed by fluctuations towards the later weeks. The dominant frequency follows a similar downward trend, suggesting a relationship between structural changes in the metal sheet and electrochemical conditions, though further analysis is needed to establish a direct correlation between mechanical degradation and corrosion progression.

At 40°C (Figure 7 (b)), a more pronounced and continuous decline in both HCP and frequency is observed, suggesting an accelerated corrosion process due to higher temperature. This reinforces the influence of temperature on corrosion rate, as increased temperature promotes electrochemical reactions. The similar trends in frequency drop and HCP further validate the capability of MEMS-SAMS in detecting and monitoring corrosion progression in reinforced concrete structures.

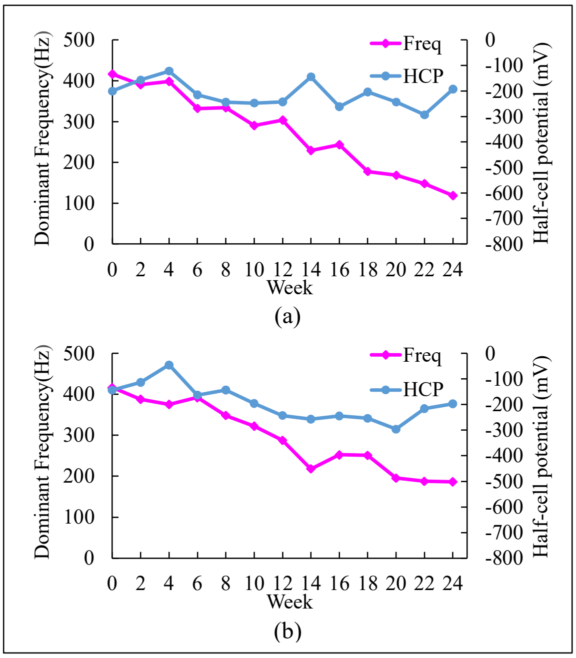

Fig. 7. Dominant frequency degradation and half-cell potential over time for deionized water solution in wet and dry conditions: (a) 20°C and (b) 40°C.

Figure 8 illustrates the dominant frequency degradation over time under 40°C submerged conditions for (a) NaCl solution and (b) deionized water, emphasizing the effectiveness of MEMS-SAMS in detecting corrosion. The results show that frequency degradation, measured by MEMS-SAMS, provides a reliable indicator of structural deterioration, with noticeable drops occurring after electrochemical changes.

In Figure 8a (NaCl solution), HCP remains well below -350 mV, indicating possible oxygen starvation. HCP initially increases at week 4 before declining, while MEMS-SAMS detects the frequency drop at week 6. The similar pattern is observed, where HCP rises at week 20, followed by the frequency drop at week 22. This demonstrates that MEMS-SAMS can effectively capture the mechanical degradation of the sacrificial anode metal sheet (SAMS), responding shortly after electrochemical changes occur.

In Figure 8b (deionized water), the trends of frequency degradation and HCP are more closely aligned, with a slight HCP increase at week 18, followed by a frequency drop at week 20. The consistent two-week delay between HCP fluctuations and MEMS-SAMS frequency drops in both submerged conditions reinforces the reliability of MEMS-SAMS as a real-time corrosion monitoring tool. By directly measuring the structural impact of corrosion, MEMS-SAMS offers an effective approach for early detection, complementing traditional electrochemical methods.

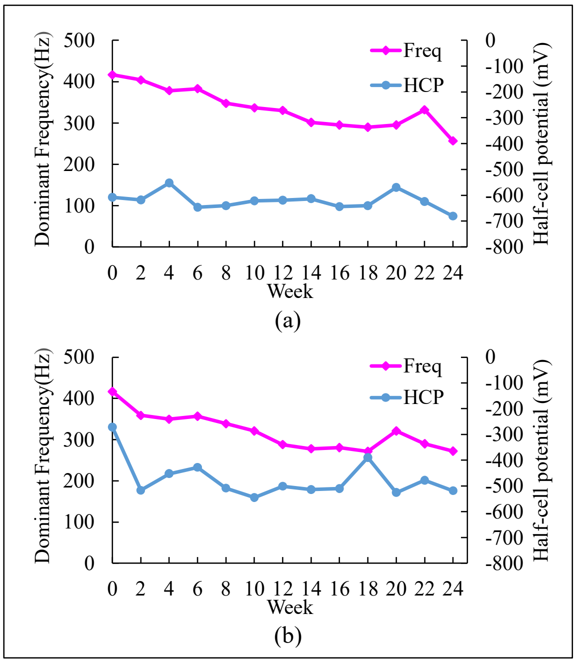

Fig. 8. Dominant frequency degradation and half-cell potential over time under 40°C submerged conditions: (a) NaCl solution and (b) deionized water.

These findings confirm that MEMS-SAMS technology can effectively predict the onset of corrosion within concrete structures by detecting frequency shifts associated with material degradation. The differences in trends between frequency degradation and HCP drop suggest that MEMS-SAMS can still provide valuable insights into corrosion severity, despite the varying patterns. By analyzing this correlation, it becomes possible to estimate the extent of damage occurring within the reinforced concrete and assess the deterioration under different environmental conditions. For early detection, MEMS-SAMS technology offers the potential to facilitate proactive maintenance planning, ensuring timely interventions before significant structural damage occurs. This capability is particularly useful in large-scale infrastructure monitoring, where identifying areas at high risk of corrosion can help prioritize repair efforts and optimize maintenance resources.

3.2 The relationship between HCP results and Power generation at target frequencies of 350 Hz and 300 Hz results

In the power generation analysis, the electrical output is derived from the dominant frequency of the SAMS. When this dominant frequency closely aligns with the MEMS device's target frequency (DTF), a significant peak in output power is observed. The magnitude of electrical power generation is further influenced by the vibration amplitude at this dominant frequency. The absolute electrical output values are calculated based on both frequency and amplitude, serving as key indicators of structural changes. However, rather than focusing on power generation itself, this output can function as a trigger for the sensor, activating a visual monitoring indicator such as LED flashing to signal potential corrosion risks. As corrosion progresses, structural degradation alters the natural frequency of the SAMS, bringing it closer to or further from the target frequency. When the natural frequency coincides with the target frequency, energy harvesting becomes most efficient, producing higher power output. This correlation between frequency shifts and power generation not only validates the effectiveness of MEMS-SAMS in detecting corrosion but also highlights its capability for real-time structural health monitoring.

Figure 9 illustrates the relationship between power generation and HCP variation over 24 weeks for NaCl solution under wet and dry conditions at (a) 20°C and (b) 40°C. The trends at DTF of 300 Hz and 350 Hz highlight the effectiveness of MEMS-SAMS in detecting corrosion. In Figure 9(a), power generation for 300 Hz appears before the first significant drop in HCP at 10 weeks. This indicates that MEMS-SAMS detects mechanical changes associated with corrosion progression before electrochemical changes become evident. HCP reaches its first lowest point before temporarily increasing, while power generation at 300 Hz has already occurred. This further demonstrates the system's ability to provide early warnings of structural changes before substantial electrochemical deterioration is detected.

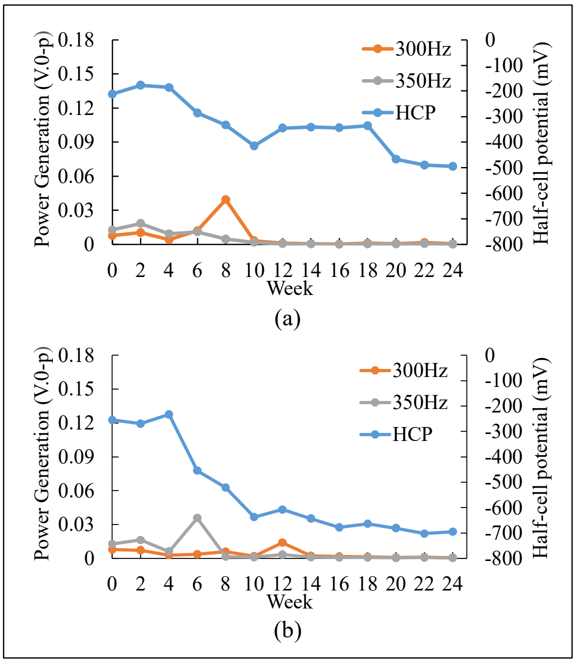

Fig. 9. Power generation at target frequencies of 350 Hz and 300 Hz and half-cell potential over time for NaCl solution in wet and dry conditions: (a) 20°C and (b) 40°C.

In Figure 9 (b), power generation for 350 Hz peaks at 6 weeks when HCP undergoes a sharp decline. As HCP continues to decrease, it reflects the ongoing corrosion of the steel reinforcement inside the concrete. The alignment between power generation and HCP reduction at elevated temperatures demonstrates that MEMS-SAMS can detect environmental changes influencing corrosion and provide timely indications of structural deterioration. While the porous nature of the concrete may facilitate chloride transport, direct chloride concentration measurements at the steel surface are needed to confirm its presence and quantify its role in corrosion progression.

Figure 10 illustrates power generation at DTF of 350 Hz and 300 Hz alongside half-cell potential (HCP) variations over time for deionized water in wet and dry conditions at (a) 20°C and (b) 40°C. Compared to NaCl exposure, deionized water induces less severe corrosion, leading to a delayed yet more pronounced power generation response. This is because lower corrosion aggressiveness preserves structural integrity longer, allowing for sustained vibrational energy harvesting.

In Figure 10 (a), the peak power of DTF at 300 Hz appears at week 12, occurring before the increase in HCP observed at week 14. This suggests the MEMS-SAMS system can detect signs of corrosion-related changes before electrochemical measurements indicate corrosion progression. Similarly, in Figure 10 (b), power generation at 350 Hz reaches a high point precisely when HCP begins to decline, demonstrating the device's capability to identify corrosion activity as it progresses. These trends highlight the effectiveness of MEMS-SAMS in tracking corrosion development across different environmental conditions, reinforcing its potential for real-time structural health monitoring.

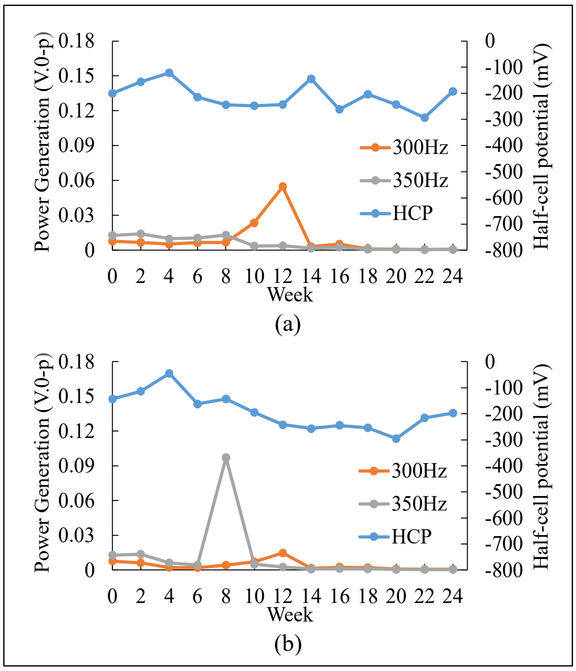

Fig. 10. Power generation at target frequencies of 350 Hz and 300 Hz and half-cell potential over time for deionized water solution in wet and dry conditions: (a) 20°C and (b) 40°C.

Figure 11 presents power generation at DTF of 350 Hz and 300 Hz alongside half-cell potential (HCP) variations over time under 40°C submerged conditions for (a) NaCl solution and (b) deionized water. Although the HCP values in submerged conditions are significantly lower than in other environments due to continuous immersion, analysing the trends reveals key insights into corrosion progression and power generation behaviour.

In Figure 11 (a), power generation at 350 Hz appears two weeks after an initial HCP drop, suggesting a delayed response in frequency degradation due to corrosion. Additionally, at week 14, power generation at 300 Hz reaches a peak just before a slight decline in HCP, reinforcing the MEMS-SAMS system's sensitivity to structural changes before electrochemical measurements fully capture corrosion activity. For Figure 11 (b), although the trend is less distinct, insights from Figure 8b help clarify the behavior. A noticeable frequency drop occurs before a slight increase in the following two weeks, aligning with a peak in power generation.

However, due to the continuous immersion, corrosion progression in submerged conditions is less severe than in wet and dry cycles, making it difficult to attribute power generation peaks solely to corrosion. To confirm the correlation, future experiments should include direct observations of rebar conditions to assess the extent of actual corrosion formation, ensuring a more comprehensive understanding of MEMS-SAMS performance in submerged environments.

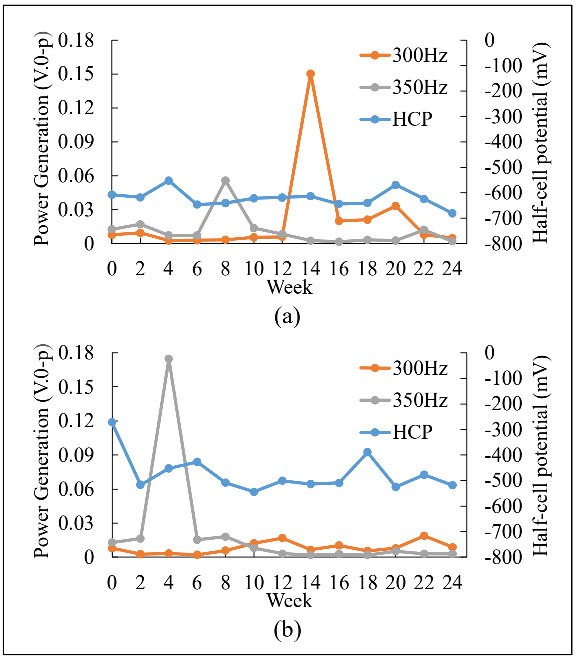

Fig. 11. Power generation at target frequencies of 350 Hz and 300 Hz and half-cell potential over time under 40°C submerged conditions: (a) NaCl solution and (b) deionized water.

In wet and dry cycles, power generation peaks closely precede or coincide with significant drops in HCP, demonstrating the MEMS-SAMS device can detect corrosion initiation at an early stage. The presence of NaCl accelerates corrosion, causing earlier and more noticeable frequency shifts, whereas deionized water leads to slower corrosion but higher power output due to less aggressive material degradation. To fully establish the link between chloride-induced corrosion and these observations, the future studies should measure chloride concentration at the steel surface. While the porous nature of the concrete likely facilitates chloride penetration, its exact concentration remains unverified in this study. In submerged conditions, the relationship between frequency shifts and HCP is less pronounced, suggesting a more stable electrochemical state under continuous immersion. However, the observed power generation peaks confirm that frequency monitoring remains a useful method for identifying early structural changes.

Ultimately, validating MEMS-SAMS through HCP measurements enhances its potential for practical application in long-term structural health monitoring. The electrical output, derived from frequency shifts and amplitude changes, serves as an indicator rather than a direct measure of corrosion severity. Instead of focusing on power generation levels, MEMS-SAMS functions as a trigger sensor, providing a visual alert, such as LED flashing when environmental conditions indicate a high risk of corrosion. By confirming its ability to detect early signs of deterioration in reinforced concrete structures, the findings of this study contribute to the development of more efficient and proactive maintenance strategies for reinforced concrete infrastructure.

Furthermore, establishing a clear relationship between HCP values and the electrical output power of MEMS-SAMS will improve the ability to distinguish between different environmental conditions affecting the concrete. Future research should focus on refining the precision of electrical power output measurements, enabling users to accurately identify specific environmental conditions influencing corrosion. This refinement will facilitate the interpretation of results and help determine the type of corrosive exposure affecting a structure, thereby supporting more effective preventive maintenance strategies.

Additionally, future studies should include field experiments to validate the practical performance of MEMS-SAMS in real-world conditions. These experiments will ensure that external factors, such as wind and traffic loads, do not interfere with the sensor's accuracy. Conducting such tests on actual bridge structures or exposed concrete elements will confirm the reliability and robustness of MEMS-SAMS technology for long-term corrosion monitoring in reinforced concrete structures.

4 Conclusion

This study demonstrates the potential of MEMS-SAMS for corrosion monitoring in reinforced concrete structures by analyzing frequency degradation and power generation alongside half-cell potential (HCP) measurements. The system effectively responds to environmental changes, capturing critical shifts in exposure conditions that influence corrosion processes. In wet and dry conditions, where corrosion progresses more aggressively, MEMS-SAMS provides an early indication of deterioration, often before significant HCP changes occur. In submerged conditions, frequency shifts and power generation are less distinct, requiring additional verification, such as direct rebar inspection, to further enhance their accuracy. To optimize MEMS-SAMS for predicting rebar corrosion, the further studies need to refine the relationship between environmental exposure and corrosion progression, considering chloride transport rates and concrete properties. By integrating frequency-based sensing with electrochemical data, MEMS-SAMS enhances early detection capabilities, supporting proactive maintenance strategies for reinforced concrete structures in various exposure conditions.

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